Association Between Time of Day of Elective General Surgery and 30-Day Mortality

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Abstract

This retrospective cohort study evaluated whether surgical start time is associated with 30-day postoperative mortality among 31,779 elective general surgery patients treated at the Cleveland Clinic from 2005 to 2010. Surgical start time was grouped into six intervals, and multivariable logistic regression adjusted for demographic, clinical, and procedural factors. Continuous covariates were assessed using LOESS and restricted cubic splines(RCS). Unadjusted mortality increased across the day (0.29% in 6–8 am vs. 0.97% in 5–7 pm), but after adjustment, start time was not associated with mortality (overall $p = 0.556$), and adjusted predicted mortality remained uniformly low (0.005%–0.043%). American Society of Anesthesiologists status(ASA) and risk stratification index(RSI) for 30-day mortality were the strongest predictors. These findings indicate that the apparent late-day increase in mortality is explained by differences in patient risk rather than surgical timing.

Keywords: surgical start time, postoperative mortality, logistic regression, restricted cubic splines

Introduction

Elective operations can be scheduled flexibly throughout the day because delaying urgent or emergency procedures can introduce substantial risks. Nonetheless, the time of day at which surgery is performed may influence postoperative outcomes. Circadian rhythm disruption during the perioperative period may reduce physiologic resilience,[1] and fatigue among surgeons and operating room staff may impair performance as the workday progresses.[2, 3, 4, 5]

However postoperative mortality is strongly shaped by patient-level risk factors. Physiologic status and comorbidity burden—captured by the ASA physical status classification, RSI(30 day mortality), and older age—are among the most consistent predictors of short-term mortality and may vary by time of day due to scheduling patterns. Evaluating whether surgical start time is associated with mortality therefore requires careful adjustment for these factors to distinguish true timing effects from underlying patient complexity.

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Research Question

Is surgical start time associated with 30-day postoperative mortality after accounting for major patient-level risk factors such as ASA status, comorbidity burden, RSI and age?

Materials and Methods

This retrospective cohort study used data from the Teaching Statistics in the Health Sciences (TSHS) repository, originally derived from Sessler et al. (2011), containing 32,001 elective general surgery patients treated at the Cleveland Clinic from 2005 to 2010.[2]

The primary outcome was 30-day postoperative mortality, and the primary exposure was the hour of surgery at case start. Covariates considered potential confounders included age, RSI, the Charlson Comorbidity Index, gender, ASA physical status, the Agency for Healthcare Research and Quality Clinical Classifications Software (AHRQ-CCS) procedure category, and baseline comorbidities (cancer, cardiovascular or cerebrovascular disease, dementia, diabetes, digestive disease, osteoarthritis, psychiatric disorder, and pulmonary disease).

Procedure types were grouped into clinically meaningful categories. Endocrine, breast, and plastic procedures, each contributing only one mortality event, were combined into an "other non-major procedures" category, resulting in five final groups: gastrointestinal and hernia, orthopedic and spine, gynecologic, urologic, and other non-major procedures.

Statistical Methods

All analyses were conducted in RStudio version 2025.09.2+418. Baseline characteristics were summarized using means and standard deviations for continuous variables and counts with percentages for categorical variables. Group differences were assessed using t tests or one-way ANOVA for continuous variables and chi-square or Fisher exact tests for categorical variables.

The association between surgical start time and 30-day mortality was evaluated using multivariable logistic regression including all prespecified covariates. The complication-specific RSI was excluded because of high correlation with the mortality-specific RSI. Clinically standard reference categories were used (e.g., 8–11 am for start time, ASA I–II for ASA status, and male sex for gender).

Continuous covariates were assessed for nonlinearity using LOESS and RCS.[6] RCS are recommended to minimize residual confounding, with 3–5 knots commonly used.[7] Candidate nonlinear variables were modeled jointly using RCS. Models with 3, 4, and 5 knots were compared against a fully linear specification, and the number of knots with the lowest Akaike Information

Table 1: Baseline Characteristics of the Study Population, Overall and by Surgical Start Time

Variable	Overall N = 31,779[1]	6-8am N = 11,117[1]	8-11am N = 7,896[1]	11-1pm N = 5,579[1]	1-3pm N = 4,664[1]	3-5pm N = 2,005[1]	5-7pm N = 518[1]	p-value[2]
Age	58 ± 15	58 ± 14	58 ± 15	58 ± 15	58 ± 15	58 ± 15	56 ± 16	0.021
Risk Stratification Index (30-day mortality)	-0.53 ± 1.04	-0.64 ± 1.04	-0.59 ± 1.05	-0.48 ± 1.03	-0.41 ± 0.99	-0.25 ± 1.02	-0.10 ± 0.96	<0.001
Charlson Comorbidity Index	1.19 ± 1.88	1.15 ± 1.79	1.13 ± 1.88	1.23 ± 1.97	1.30 ± 1.94	1.26 ± 1.91	1.21 ± 1.99	<0.001
Gender								<0.001
Male	17,115 (54%)	5,559 (50%)	4,611 (58%)	3,088 (55%)	2,469 (53%)	1,105 (55%)	283 (55%)	
Female	14,664 (46%)	5,558 (50%)	3,285 (42%)	2,491 (45%)	2,195 (47%)	900 (45%)	235 (45%)	
ASA Physical Status								<0.001
I-II	17,079 (54%)	6,284 (57%)	4,250 (54%)	2,874 (52%)	2,369 (51%)	1,036 (52%)	266 (51%)	
III	13,647 (43%)	4,574 (41%)	3,381 (43%)	2,481 (44%)	2,120 (45%)	871 (43%)	220 (42%)	
IV-VI	1,053 (3.3%)	259 (2.3%)	265 (3.4%)	224 (4.0%)	175 (3.8%)	98 (4.9%)	32 (6.2%)	
Procedure								<0.001
Other non-major procedures	4,291 (14%)	1,144 (10%)	1,249 (16%)	863 (15%)	634 (14%)	318 (16%)	83 (16%)	
Gynecologic	4,512 (14%)	1,326 (12%)	1,217 (15%)	824 (15%)	735 (16%)	317 (16%)	93 (18%)	
Urologic	6,639 (21%)	2,872 (26%)	1,266 (16%)	891 (16%)	1,132 (24%)	400 (20%)	78 (15%)	
Orthopedic & Spine	10,848 (34%)	4,286 (39%)	2,718 (34%)	1,883 (34%)	1,326 (28%)	528 (26%)	107 (21%)	
Gastrointestinal & Hernia	5,489 (17%)	1,489 (13%)	1,446 (18%)	1,118 (20%)	837 (18%)	442 (22%)	157 (30%)	
Cancer	10,942 (34%)	3,900 (35%)	2,567 (33%)	1,798 (32%)	1,802 (39%)	727 (36%)	148 (29%)	<0.001
Cardiovascular/Cerebrovascular disease	16,162 (51%)	5,659 (51%)	3,980 (50%)	2,860 (51%)	2,390 (51%)	1,022 (51%)	251 (48%)	0.8
Dementia	241 (0.8%)	58 (0.5%)	63 (0.8%)	45 (0.8%)	55 (1.2%)	16 (0.8%)	4 (0.8%)	0.002
Diabetes	4,163 (13%)	1,474 (13%)	1,006 (13%)	776 (14%)	605 (13%)	247 (12%)	55 (11%)	0.14
Digestive disease	7,020 (22%)	2,422 (22%)	1,765 (22%)	1,252 (22%)	1,034 (22%)	441 (22%)	106 (20%)	0.8
Osteoarthritis	5,711 (18%)	2,167 (19%)	1,684 (21%)	1,006 (18%)	622 (13%)	199 (9.9%)	33 (6.4%)	<0.001
Psychiatric disorder	2,904 (9.1%)	962 (8.7%)	728 (9.2%)	577 (10%)	410 (8.8%)	186 (9.3%)	41 (7.9%)	0.012
Pulmonary disease	3,467 (11%)	1,150 (10%)	865 (11%)	633 (11%)	517 (11%)	241 (12%)	61 (12%)	0.2
30-day Mortality	138 (0.4%)	32 (0.3%)	25 (0.3%)	36 (0.6%)	21 (0.5%)	19 (0.9%)	5 (1.0%)	<0.001

[1] Mean ± SD; n (%)

[2] One-way analysis of means; Pearson's Chi-squared test; Fisher's Exact Test for Count Data with simulated p-value (based on 10000 replicates)

Criterion (AIC) was selected. Using this optimal spline structure, each variable was then tested for nonlinearity using Wald χ^2 statistics, and spline terms were retained only when the nonlinearity was statistically significant.

For the primary exposure (hour of surgery), both spline-based and categorical parameterizations were considered. Categorical modeling was prespecified using six clinically meaningful intervals (6–8 am, 8–11 am, 11 am–1 pm, 1–3 pm, 3–5 pm, and 5–7 pm), following recommendations to use ≤4-hour time blocks aligned with circadian patterns.[1] This approach also allowed comparison with prior studies, such as Sessler et al., which treated hour as a continuous exposure.

Associations with mortality were reported as odds ratios (ORs) with 95% confidence intervals (CIs). For the primary exposure, predicted 30-day mortality percentages from unadjusted and adjusted models were additionally examined.

Results

The final analytic cohort included 31,779 adult patients (mean±SD age, 58±15 years; 14,664 [46%] female). Patients younger than 18 years (n=212) were excluded because the original study population and associated risk indices were developed for adult surgery, along with 2 patients missing gender and 8 missing ASA physical status. Table 1 summarizes baseline characteristics by surgical start time. Surgery hour was categorized into six intervals (6–8 am, 8–11 am, 11–1 pm, 1–3 pm, 3–5 pm, and 5–7 pm). Most operations occurred during the morning hours (6–11 am), with substantially fewer surgeries performed later in the day(5-7 pm), reflecting typical scheduling and operating room workflows. Significant differences across groups were observed in 30-day mortality and several covariates, including age, RSI, Charlson Comorbidity Index, gender, ASA physical status, procedure type, and selected comorbidities (cancer, dementia, osteoarthritis, and psychiatric disorder).

A LOESS assessment suggested potential nonlinear associations for age, RSI, and hour of surgery. When these variables were modeled jointly using restricted cubic splines, a 3-knot specification provided the best fit (AIC=1242), compared with 4 knots (AIC=1249), 5 knots (AIC=1244), or a linear specification (AIC=1247). Subsequent tests showed strong evidence of nonlinearity for RSI ($p<0.001$), but not for age ($p=0.324$) or hour of surgery ($p=0.999$). Accordingly, RSI was modeled using a 3-knot spline, whereas age was modeled linearly and hour of surgery was modeled categorically.

In adjusted logistic regression analyses, RSI was strongly associated with 30-day mortality (overall $p<0.0001$). The spline demonstrated a sharply increasing risk beginning around RSI≈2, with predicted mortality exceeding 10% at the highest RSI levels (Figure 1).

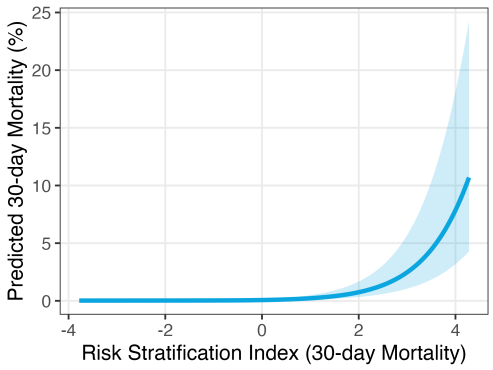

Figure 1: Adjusted Spline Relationship Between the Risk Stratification Index and Predicted 30-Day Postoperative Mortality

Hour of surgery was not associated with postoperative mortality in adjusted analyses (overall p=0.556). Compared with procedures performed between 8–11 am, surgeries conducted during later intervals (11 am–1 pm, 1–3 pm, 3–5 pm, and 5–7 pm) did not exhibit significantly different odds of death (Figure 3). In unadjusted analyses, predicted 30-day mortality increased throughout the day, from approximately 0.29% during 6–8 am to 0.97% during 5–7 pm (Figure 2). After adjustment for demographic, clinical, and procedural factors, these differences were markedly attenuated. Adjusted predicted mortality remained uniformly low across all hour categories, ranging from 0.005% to 0.043%, with no meaningful variation by time of day. The highest adjusted estimate occurred during 8–11 am (0.043%), while all other intervals ranged from 0.005% to 0.007%, indicating no clinically meaningful association between surgery hour and 30-day mortality.

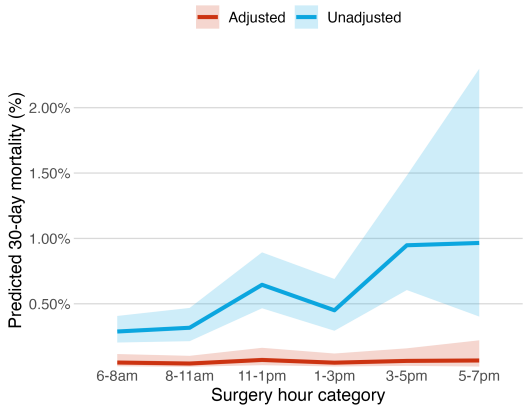

Figure 2: Comparison of Unadjusted and Adjusted Predicted 30-Day Postoperative Mortality by Surgical Start Time

ASA status was the strongest independent predictor of postoperative mortality. Compared with patients classified as ASA I–II, those with ASA IV–VI had substantially higher odds of death (OR, 9.84 [95% CI, 4.69–20.63]), as did those with ASA III (OR, 3.43 [95% CI, 1.77–6.65]). Osteoarthritis (OR, 2.24 [95% CI, 1.10–4.55]) and female sex (OR, 1.52 [95% CI, 1.03–2.24]) were also associated with increased mortality. Age was positively associated with risk (per-year OR, 1.02 [95% CI, 1.01–1.04]). No significant associations were observed for most other comorbidities or procedure categories.

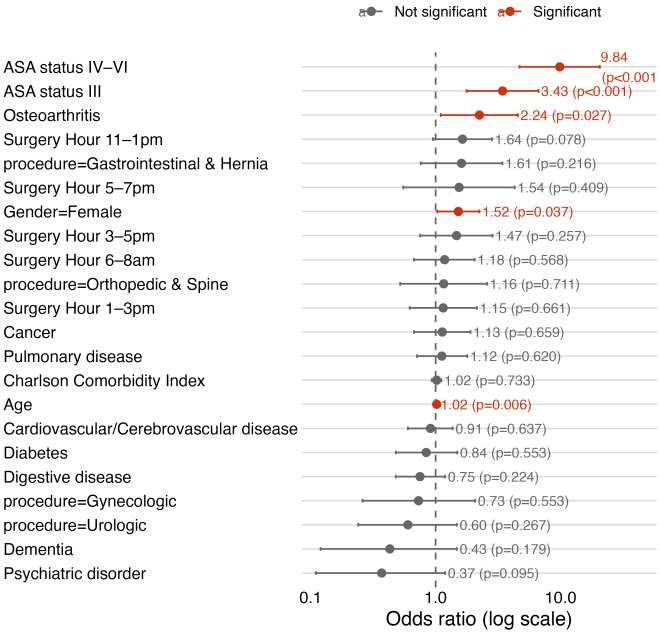

Figure 3: Adjusted Odds Ratios and 95% Confidence Intervals for Predictors of 30-Day Postoperative Mortality

Conclusion

After accounting for patient characteristics, comorbidities, and procedure type, the hour of surgery showed no meaningful association with 30-day postoperative mortality. The apparent increase in unadjusted mortality later in the day was explained by differences in patient risk profiles rather than timing itself. In contrast, ASA status and overall physiologic burden, as measured by the Risk Stratification Index, were the strongest predictors of mortality.

These findings suggest that, within this surgical environment, short-term mortality risk is driven primarily by patient health status rather than the specific time surgery begins. Limitations include the observational design, potential residual confounding, incomplete capture of operational factors such as staffing patterns or fatigue, and the use of data from a single health system, which may affect generalizability.

References

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Code Appendix

2025-12-03

```r
## ================================
## 0. Load needed packages and data
## ================================
# set up the environment
setwd("~/Dropbox/School/CU/fall 2025/BIOS 6618 adv biostatistical method/final project/surgery_timing_ou
# Helper function: install if not already available
load_or_install <- function(pkg){
  if (!requireNamespace(pkg, quietly = TRUE)) {
    install.packages(pkg)
  }
  library(pkg, character.only = TRUE)
}

# List of needed packages
packages <- c("dplyr", "ggplot2",  "kableExtra", "webshot2","magick","knitr","tidyverse",
              "tidyr","ggh4x","GGally","visdat","car","fastDummies","naniar",
              "rms","Hmisc","patchwork","forcats","xfun","gtsummary","gt")
# Load or install each
lapply(packages, load_or_install)
data <- read.csv("./data/Surgery_Timing.csv")

##############################################################
## 1. Define variable groups (by role & type)
##############################################################

# Primary exposure & outcomes
var_exposure   <- "hour"
var_outcome   <- "mort30"

# Comorbidity indicators (0/1)
vars_comorbidity <- c(
  "baseline_cancer",
  "baseline_cvd",
  "baseline_dementia",
  "baseline_diabetes",
  "baseline_digestive",
  "baseline_osteoart",
  "baseline_psych",
  "baseline_pulmonary"
)

# Continuous numeric variables for EDA
vars_continuous <- c(
  "age",
```

```r
  "bmi",
  "baseline_charlson",
  "mortality_rsi",
  "complication_rsi",
  "ccsMort30Rate",
  "ccsComplicationRate",
  "hour"
)

# Binary variables (0/1)
vars_binary <- c(
  vars_comorbidity,
  var_outcome
)

# Categorical variables
vars_categorical <- c(
  "gender",
  "race",
  "asa_status",
  "ahrq_ccs",
  "dow",
  "month",
  "moonphase"
)
n_cols <- ncol(data)
print(length(vars_categorical) + length(vars_binary) + length(vars_continuous) == n_cols)  # should equ
vars_categorical_binary <- c(vars_categorical, vars_binary)
##############################################################
## 2. Type conversion
##############################################################

data <- data %>%
  # Convert binary variables to factors
  mutate(across(all_of(vars_binary), ~ as.numeric(.))) %>%
  # Convert categorical variables to factors
  mutate(across(all_of(vars_categorical), ~ as.factor(.))) %>%
  # convert continuous variables to numeric
  mutate(across(all_of(vars_continuous), ~ as.numeric(.))
  )
```

```r
cat("=== Structure ===\n")
str(data)
summary(data)

##############################################################
# logic cleaning
##############################################################
# keep age >= 18
orin_dim <- dim(data)
# find mort30 for age < 18
mort_young <- data %>%
  dplyr::filter(age < 18) %>%
```

```r
  dplyr::select(mort30)
cat("30-day mortality for patients age < 18:\n")
print(table(mort_young$mort30))
print(paste0("Original data dimensions: ", orin_dim[1], " rows, ", orin_dim[2], " columns"))
data <- data %>%
  filter(age >= 18)
cat("Data dimensions after filtering age >= 18:\n")
print(dim(data))
removed_rows <- orin_dim[1] - nrow(data)
cat(paste0("Number of rows removed due to age < 18: ", removed_rows, "\n"))
# check age summary
summary(data$age)


##############################################################
## multicollinearity check for continuous variables
##############################################################
# calculate vif for continuous variables
vif_data <- data %>%
  dplyr::select(all_of(c(vars_continuous, "mort30"))) %>%
  na.omit()
vif_model <- lm(mort30 ~ ., data = vif_data)
vif_values <- vif(vif_model)
vif_results <- data.frame(
  variable = names(vif_values),
  vif      = as.numeric(vif_values),
  stringsAsFactors = FALSE
)
print(vif_results)
# return variables with vif > 5
high_vif_vars <- vif_results$variable[vif_results$vif > 10]
cat("Continuous variables with high VIF (> 5):\n")
print(high_vif_vars)

# calculate correlation matrix for continuous variables
cor_data <- data %>%
  dplyr::select(all_of(vars_continuous)) %>%
  na.omit()
cor_matrix <- cor(cor_data)
print(cor_matrix)
# return variable pairs with high correlation (> 0.7)
# remove duplicate pairs and self-correlation
high_cor_results <- data.frame(
  variable1 = character(),
  variable2 = character(),
  correlation = numeric(),
  stringsAsFactors = FALSE
)
for (i in 1:(ncol(cor_matrix) - 1)) {
  for (j in (i + 1):ncol(cor_matrix)) {
    cor_value <- cor_matrix[i, j]
    if (abs(cor_value) > 0.7) {
      high_cor_results <- rbind(
```

```
        high_cor_results,
        data.frame(
          variable1   = colnames(cor_matrix)[i],
          variable2   = colnames(cor_matrix)[j],
          correlation = cor_value,
          stringsAsFactors = FALSE
        )
      )
    }
  }
}
print(high_cor_results)
# Based on the results, consider removing one variable from each highly correlated pair
numeric_vars_remove <- c("bmi","complication_rsi","ccsComplicationRate")
# final continuous variables to consider
numeric_vars_consider <- setdiff(vars_continuous, numeric_vars_remove)
cat("Variable pairs with high correlation (> 0.7):\n")
print(high_cor_results)
cat("Removing the following continuous variables due to high multicollinearity:\n")
print(numeric_vars_remove)
cat("Continuous variables to consider after removing multicollinear variables:\n")
print(numeric_vars_consider)

##############################################################
#missing value imputation
##############################################################
# check missingness
missing_summary <- data.frame(
  variable       = character(),
  missing_count  = numeric(),
  missing_percent = numeric(),
  stringsAsFactors = FALSE
)
for (v in colnames(data)) {
  missing_count <- sum(is.na(data[[v]]))
  missing_percent <- missing_count / nrow(data) * 100
  missing_summary <- rbind(
    missing_summary,
    data.frame(
      variable        = v,
      missing_count   = missing_count,
      missing_percent = missing_percent,
      stringsAsFactors = FALSE
    )
  )
}
# check missingness by mort30 and pivot by mort30
missing_by_mort30 <- data %>%
  group_by(mort30) %>%
  summarise(across(everything(), ~ sum(is.na(.)))) %>%
  tidyr::pivot_longer(
    cols = -mort30,
    names_to = "variable",
```

```r
    values_to = "missing_count"
  ) %>%
  tidyr::pivot_wider(
    names_from = mort30,
    values_from = missing_count,
    names_prefix = "mort30_"
  )
missing_summary <- missing_summary %>%
  left_join(missing_by_mort30, by = "variable")
print(missing_summary)

##############################################
# finalize variable lists for modeling
##############################################
vars_to_check <- c("age", "gender", "asa_status")
vars_to_drop  <- c("bmi", "race",
                   "ccsComplicationRate","complication_rsi"
                   )
cat("Variables to drop due to  high correlation or high missingness:\n")
print(vars_to_drop)
data_clean <- data %>%
  drop_na(all_of(vars_to_check)) %>%
  dplyr::select(-all_of(vars_to_drop))
cat("Data dimensions after dropping missing values in key variables and unnecessary variables:\n")
print(dim(data_clean))
print(length(vars_to_drop))
length(vars_to_drop) + ncol(data_clean) == ncol(data)  # should equal ncol(data)
# final variable lists
final_vars_categorical <- setdiff(vars_categorical, vars_to_drop)
final_vars_binary      <- setdiff(vars_binary, vars_to_drop)
final_vars_continuous  <- numeric_vars_consider
cat("Final categorical variables:\n")
print(final_vars_categorical)
cat("Final binary variables:\n")
print(final_vars_binary)
cat("Final continuous variables:\n")
print(final_vars_continuous)
# check missingness again
final_missing_summary <- data.frame(
  variable       = character(),
  missing_count  = numeric(),
  missing_percent = numeric(),
  stringsAsFactors = FALSE
)
for (v in colnames(data_clean)) {
  missing_count <- sum(is.na(data_clean[[v]]))
  missing_percent <- missing_count / nrow(data_clean) * 100
  final_missing_summary <- rbind(
    final_missing_summary,
    data.frame(
      variable        = v,
      missing_count   = missing_count,
      missing_percent = missing_percent,
```

```
      stringsAsFactors = FALSE
    )
  )
}
print(final_missing_summary)

##############################################################
# combine ahrq_ccs levels for modeling
##############################################################
data_clean <- data_clean %>%
  mutate(
    procedure_group = case_when(
      ahrq_ccs %in% c("Colorectal resection",
                      "Gastrectomy; partial and total",
                      "Small bowel resection",
                      "Inguinal and femoral hernia repair",
                      "Other hernia repair") ~ "Gastrointestinal & Hernia",

      ahrq_ccs %in% c("Lumpectomy; quadrantectomy of breast",
                      "Mastectomy") ~ "Breast",

      ahrq_ccs %in% c("Arthroplasty knee",
                      "Hip replacement; total and partial",
                      "Laminectomy; excision intervertebral disc",
                      "Spinal fusion") ~ "Orthopedic & Spine",

      ahrq_ccs %in% c("Hysterectomy; abdominal and vaginal",
                      "Oophorectomy; unilateral and bilateral",
                      "Other excision of cervix and uterus",
                      "Repair of cystocele and rectocele; obliteration of vaginal vault") ~ "Gynecologic

      ahrq_ccs %in% c("Endoscopy and endoscopic biopsy of the urinary tract",
                      "Nephrectomy; partial or complete",
                      "Open prostatectomy",
                      "Transurethral resection of prostate (TURP)") ~ "Urologic",

      ahrq_ccs %in% c("Thyroidectomy; partial or complete") ~ "Endocrine",

      ahrq_ccs %in% c("Plastic procedures on nose") ~ "Plastic",

      TRUE ~ "Other non-major procedures"
    )
  )
# check procedure group by original ahrq_ccs
print(unique(data_clean$procedure_group))
procedure_check <- data_clean %>%
  group_by(procedure_group,ahrq_ccs) %>%
  summarise(count = n()) %>%
  arrange(procedure_group)
print(procedure_check, n=length(unique(data_clean$ahrq_ccs)))

#plot histogram of mort30 by procedure_group
ggplot(data_clean, aes(x=procedure_group, fill=factor(mort30))) +
```

```r
  geom_bar(position="dodge") +
  labs(title="30-day Mortality by Procedure Group", x="Procedure Group", y="Count", fill="30-day Mortali
  theme_minimal()
```

```r
# summarize mort30 by procedure_group
# check sample size (n<100?) and death(n<10?) in each procedure_group
group_summary <- data_clean %>%
  group_by(procedure_group) %>%
  summarise(
    n          = n(),
    deaths     = sum(mort30, na.rm = TRUE),
    mort_rate  = mean(mort30, na.rm = TRUE),
    .groups    = "drop"
  ) %>%
  arrange(desc(n))

group_summary

# check heterogeneity within each group
# within the group, are mortality rates similar across different ahrq_ccs?
within_group <- data_clean %>%
  group_by(procedure_group, ahrq_ccs) %>%
  summarise(
    n          = n(),
    deaths     = sum(mort30, na.rm = TRUE),
    mort_rate  = mean(mort30, na.rm = TRUE),
    .groups    = "drop"
  ) %>%
  arrange(procedure_group, desc(n))

print(within_group, n=27)

ggplot(group_summary, aes(x = procedure_group, y = mort_rate)) +
  geom_col() +
  geom_text(
    aes(label = sprintf("%.2f%%", mort_rate * 100)),
    vjust = -0.3,
    size  = 3
  ) +
  labs(
    title = "30-day mortality rate by procedure group",
    x     = "Procedure group",
    y     = "Mortality rate"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

```r
# combine group with < 6 deaths
data_clean <- data_clean %>%
  mutate(
    procedure = ifelse(
      procedure_group %in% group_summary$procedure_group[group_summary$deaths < 6],
      "Other non-major procedures",
```

```r
      procedure_group
    )
  )
# check new procedure group summary
new_group_summary <- data_clean %>%
  group_by(procedure) %>%
  summarise(
    n          = n(),
    deaths     = sum(mort30, na.rm = TRUE),
    mort_rate  = mean(mort30, na.rm = TRUE),
    .groups    = "drop"
  ) %>%
  arrange(desc(n))
print(new_group_summary)

#check finalized group, procedure group and ahrq_ccs
final_procedure_check <- data_clean %>%
  group_by(procedure, procedure_group,ahrq_ccs) %>%
  summarise(count = n()) %>%
  arrange(procedure)
print(final_procedure_check, n=length(unique(data_clean$ahrq_ccs)))

# replace ahrq_ccs with procedure_group2
final_vars_categorical <- setdiff(final_vars_categorical, "ahrq_ccs")
final_vars_categorical <- c(final_vars_categorical, "procedure")
print(final_vars_categorical)
# check unique levels in each categorical variable
for (v in final_vars_categorical) {
  cat("Variable:", v, "\n")
  print(unique(data_clean[[v]]))
}
data_clean <- data_clean %>%
  mutate(
    procedure = factor(
      procedure,
      levels = c(
        "Other non-major procedures",  # reference
        "Gynecologic",
        "Urologic",
        "Orthopedic & Spine",
        "Gastrointestinal & Hernia"
      )
    ),
    dow = factor(
      dow,
      levels=c("1","2","3","4","5")
    ),
    month = factor(
      month,
      levels=c("1","2","3","4","5","6","7","8","9","10","11","12")
    )
  )
# drop original procedure_group and ahrq_ccs
```

```r
data_to_model <- data_clean %>%
  dplyr::select(-c(procedure_group, ahrq_ccs, "complication","moonphase","dow","month","ccsMort30Rate"))
str(data_to_model)

# -------------------------
# CREATE HOUR CATEGORIES
# -------------------------
data_to_model$hour_cat <- cut(
  data_to_model$hour,
  breaks = c(6, 8, 11, 13, 15, 17, 19),
  right = FALSE,  # intervals left-inclusive
  include.lowest = FALSE,
  labels = c("6-8am", "8-11am", "11-1pm", "1-3pm", "3-5pm", "5-7pm")
)

# check distribution
table(data_to_model$hour_cat, useNA = "ifany")
#return row for na hour_cat
data_to_model[is.na(data_to_model$hour_cat), ]
#if hour=19 set it to "5-7"
data_to_model$hour_cat[is.na(data_to_model$hour_cat) & data_to_model$hour == 19] <- "5-7pm"
table(data_to_model$hour_cat, useNA = "ifany")

##################################################
# Plot baseline characteristics table
##################################################
## ---------------------------------------------------
## 0. Pre-processing: factors, labels, 0/1 → No/Yes
## ---------------------------------------------------

# set your hour_cat levels if you haven't already
data_to_plot <- data_to_model %>%
  mutate(
    # gender: 0 = Male, 1 = Female
    gender = factor(ifelse(gender == 1, "Female", "Male"),
                    levels = c("Male", "Female")),
    mort30 = factor(ifelse(mort30 == 1, "Yes", "No"),
                    levels = c("No", "Yes")),
    asa_status = factor(asa_status,
                        levels = c("1", "2", "3"),
                        labels = c(
                          "I-II",
                          "III",
                          "IV-VI"
                        )
    )
  )

# list of binary baseline comorbidity variables that are 0/1
comorb_vars <- c(
  "baseline_cancer", "baseline_cvd", "baseline_dementia",
  "baseline_diabetes", "baseline_digestive",
  "baseline_osteoart", "baseline_psych", "baseline_pulmonary"
```

```r
)

# recode all of them 0/1 -> No/Yes
data_to_plot <- data_to_plot %>%
  mutate(
    across(all_of(comorb_vars),
           ~ factor(ifelse(. == 1, "Yes", "No"),
                    levels = c("No", "Yes")))
  )

## ---------------------------------------------------
## 1. Variable sets & ordering
## ---------------------------------------------------

cont_vars <- c("age", "mortality_rsi", "baseline_charlson")

cat_vars <- names(data_to_plot)[
  !(names(data_to_plot) %in% c(cont_vars,  "hour", "hour_cat"))
]

# put main variables first, then comorbidities, then any other cats
main_cat <- c("gender", "asa_status","procedure")
other_cat <- setdiff(cat_vars, c(main_cat, comorb_vars))

var_order <- c(
  "age",
  "mortality_rsi",
  "baseline_charlson",
  main_cat,
  comorb_vars,
  other_cat
)
var_order
## ---------------------------------------------------
## 2. gtsummary baseline table
## ---------------------------------------------------

tbl_baseline <-
  data_to_plot %>%
  dplyr::select(hour_cat, all_of(var_order)) %>%
  tbl_summary(
    by = hour_cat,
    statistic = list(
      all_continuous()  ~ "{mean} ± {sd}",
      all_categorical() ~ "{n} ({p}%)"
    ),
    missing = "no",
    label = list(
      age              ~ "Age",
      mortality_rsi    ~ "Risk Stratification Index (30-day mortality)",
      baseline_charlson~ "Charlson Comorbidity Index",
      gender           ~ "Gender",
      procedure        ~ "Procedure",
```

```r
      asa_status       ~ "ASA Physical Status",
      # comorbidities
      baseline_cancer      ~ "Cancer",
      baseline_cvd         ~ "Cardiovascular/Cerebrovascular disease",
      baseline_dementia    ~ "Dementia",
      baseline_diabetes    ~ "Diabetes",
      baseline_digestive   ~ "Digestive disease",
      baseline_osteoart    ~ "Osteoarthritis",
      baseline_psych       ~ "Psychiatric disorder",
      baseline_pulmonary   ~ "Pulmonary disease",
      mort30               ~ "30-day Mortality"
    )
  ) %>%
  add_overall(last = FALSE) %>%
  add_p(
    test = list(
      all_continuous()  ~ "oneway.test",
      all_categorical() ~ "chisq.test",
      baseline_dementia ~ "fisher.test",
      mort30            ~ "fisher.test"
    ),
    test.args = list(
      all_continuous()  ~ list(var.equal = TRUE),
      baseline_dementia ~ list(simulate.p.value = TRUE, B = 10000),
      mort30            ~ list(simulate.p.value = TRUE, B = 10000)
    )
  ) %>%
  modify_header(label ~ "**Variable**") %>%
  bold_labels()

## ---------------------------------------------------
## 3. Convert to gt and save PNG
##    (wide but not too tall)
## ---------------------------------------------------

gt_tbl <- as_gt(tbl_baseline) %>%
  gt::tab_style(
    style = gt::cell_fill(color = "#f7f7ec"),
    locations = gt::cells_body()
  ) %>%
  gt::tab_style(
    style = gt::cell_fill(color = "#f7f7ec"),
    locations = gt::cells_column_labels()
  ) %>%
  gt::opt_row_striping(
    row_striping = TRUE
  ) %>%
  gt::tab_options(
    table.font.size = 9,
    data_row.padding = gt::px(1),
    heading.padding = gt::px(2)
  )
gt_tbl
```

```r
##################################################
#Modeling code here
##################################################
##################################
#load data
##################################
data <- data_to_model
data$hour_cat <- relevel(data$hour_cat, ref = "8-11am")
str(data)
# check missingness for data
miss_summary <- naniar::miss_var_summary(data)
print(miss_summary)
names(data)
dim(data)

##################################
# build models
##################################
# set up
vars_spline <- c("age", "mortality_rsi")
var_exposure_cat <- "hour_cat"
dd <- datadist(data)
options(datadist = "dd")

# automatically define other covariates
other_covariates <- setdiff(
  names(data),
  c(var_outcome, var_exposure, vars_spline, var_exposure_cat)
)
print(other_covariates)
print(length(other_covariates))

# construct base rhs
base_rhs <- paste(
  "rcs(age, K) + rcs(mortality_rsi, K) + rcs(hour, K)",
  if (length(other_covariates) > 0)
    paste("+", paste(other_covariates, collapse = " + "))
  else ""
)

# fit models with different K
fit_k <- list()
AIC_k <- numeric()

for (K in c(0, 3, 4,5)) {
  if (K == 0) {
    # k = 0 = linear model: no rcs(), just linear terms
    f <- lrm(
      as.formula(
        paste("mort30 ~ age + mortality_rsi + hour",
              if (length(other_covariates) > 0)
                paste("+", paste(other_covariates, collapse = " + "))
              else "")
```

```r
      ),
      data = data
    )
  } else {
    f <- lrm(
      as.formula(paste("mort30 ~", gsub("K", K, base_rhs))),
      data = data
    )
  }
  fit_k[[as.character(K)]] <- f
  AIC_k[as.character(K)] <- round(AIC(f),0)
}

AIC_k
# choose the k with lowest AIC
best_k <- as.numeric(names(which.min(AIC_k)))
print(paste("Best K:", best_k))
best_model <- fit_k[[as.character(best_k)]]

# within best k, check which variables are truely nonlinear
#if nonlinear has small p keep spline, if p large change to linear
anova(best_model)

# -------------------------
# FIT MODEL WITH HOUR AS CATEGORICAL VARIABLE
# -------------------------
final_model_rhs <- as.formula(
  paste("mort30 ~ age + rcs(mortality_rsi,3) + hour_cat",
        if (length(other_covariates) > 0)
          paste("+", paste(other_covariates, collapse = " + "))
        else "")
)
fit_final_basic <- lrm(
  final_model_rhs,
  data = data
)
anova(fit_final_basic)

#check AIC for all models
c(
  AIC_k,
  final=round(AIC(fit_final_basic),0)
)

# unadjusted effect of hour
fit_unadjusted <- lrm(
  mort30 ~ hour_cat,
  data = data
)


#####################################################
# plots for report purposes
```

```r
#####################################################
# this object contains coeffients, se, z statistic, wald p-values, confidence intervals
## 1) Coefficients (betas)
co <- coef(fit_final_basic)

## 2) Variance-covariance matrix and SEs
vc <- vcov(fit_final_basic)
se <- sqrt(diag(vc))

## 3) Wald Z-statistics and p-values
z  <- co / se
p  <- 2 * pnorm(-abs(z))

## 4) Wald 95% CI for betas
ci_beta <- confint.default(fit_final_basic)  # works for lrm objects

# -------------- VARIABLE LABELS & LEVEL LABELS -------------------

## 5) Put everything together on OR scale
options(scipen = 999)   # prevents scientific notation
or_table <- data.frame(
  term    = names(co),
  beta    = round(co, 3),
  se      = round(se, 3),
  z       = round(z, 3),
  p_value = ifelse(p < 0.001, "<0.001", round(p, 3)),
  OR      = round(exp(co), 2),
  lower95 = round(exp(ci_beta[, 1]), 2),
  upper95 = round(exp(ci_beta[, 2]), 2),
  row.names = NULL,
  check.names = FALSE
)
or_table <- or_table %>%
  mutate(
    clean_term = dplyr::recode(term,

                               # continuous variables
                               "age"               = "Age",
                               "mortality_rsi"     = "Risk Stratification Index (30-day mortality)",
                               "baseline_charlson" = "Charlson Comorbidity Index",

                               # gender (binary)
                               "gender=1" = "Gender=Female",

                               # ASA levels
                               "asa_status=2" = "ASA status III",
                               "asa_status=3" = "ASA status IV-VI",

                               # Procedure categories
                               "procedure=GI"     = "Gastrointestinal & Hernia (vs other)",
                               "procedure=Ortho"  = "Orthopedic & Spine (vs other)",
                               "procedure=Gyn"    = "Gynecologic (vs other)",
                               "procedure=Uro"    = "Urologic (vs other)",
```

```r
                                # Binary comorbidities (0/1)
                                "baseline_cancer"    = "Cancer",
                                "baseline_cvd"       = "Cardiovascular/Cerebrovascular disease",
                                "baseline_dementia"  = "Dementia",
                                "baseline_diabetes"  = "Diabetes",
                                "baseline_digestive" = "Digestive disease",
                                "baseline_osteoart"  = "Osteoarthritis",
                                "baseline_psych"     = "Psychiatric disorder",
                                "baseline_pulmonary" = "Pulmonary disease",

                                # Hour categories
                                "hour_cat=8-11am"   = "Surgery Hour 8-11am",
                                "hour_cat=11-1pm" = "Surgery Hour 11-1pm",
                                "hour_cat=1-3pm"    = "Surgery Hour 1-3pm",
                                "hour_cat=3-5pm"    = "Surgery Hour 3-5pm",
                                "hour_cat=5-7pm"    = "Surgery Hour 5-7pm",
                                "hour_cat=6-8am"   = "Surgery Hour 6-8am",

                                .default = term    # fallback preserves anything unmapped
    )
  )


#significant variables at alpha = 0.05
or_table %>%
  filter(p_value != "<0.001" & p_value < 0.05 | p_value == "<0.001")

# 1. Start from your existing or_table
# or_table has: term, OR, lower95, upper95, p_value, etc.

plot_df <- or_table %>%
  # remove vector of terms don't want to plot
  filter(!term %in% c("Intercept","mortality_rsi","mortality_rsi'")) %>%
  filter(!is.na(OR)) %>%
  mutate(
    # numeric p-value
    p_num = 2 * pnorm(-abs(beta / se)),

    # significance flag
    signif = p_num < 0.05 &
      ((lower95 > 1) | (upper95 < 1)),

    signif_group = ifelse(
      signif,
      "Significant",
      "Not significant"
    ),

    # formatted p-value
    label_p = ifelse(
      p_num < 0.001,
      "p<0.001",
```

```r
      paste0("p=", sprintf("%.3f", p_num))
    ),

    # text label: ASA IV-VI gets 2 lines, others stay 1 line
    label = if_else(
      clean_term == "ASA status IV-VI",
      paste0(sprintf("%.2f", OR), "\n(", label_p, ")"),
      paste0(sprintf("%.2f", OR), " (", label_p, ")")
    )
  ) %>%
  arrange(OR) %>%
  mutate(clean_term = factor(clean_term, levels = clean_term))

plot_df <- plot_df %>%
  mutate(
    OR_text = sprintf("%.2f", OR),
    p_text  = paste0("(", label_p, ")"),
    v_OR = if_else(clean_term == "ASA status IV-VI",  0.3, 0),
    v_p  = if_else(clean_term == "ASA status IV-VI", -0.3, 0)
  )


ggplot(plot_df, aes(x = OR, y = clean_term, color = signif_group)) +
  geom_vline(xintercept = 1, linetype = "dashed", color = "grey40") +

  geom_errorbarh(aes(xmin = lower95, xmax = upper95), height = 0.15) +
  geom_point(size = 2) +

  geom_text(
    aes(x = upper95, label = label),
    hjust = -0.03,
    size = 3
  ) +

  scale_x_log10(expand = expansion(mult = c(0.05, 0.20))) +

  scale_color_manual(
    name = NULL,
    values = c(
      "Significant" = "#CC3311",
      "Not significant"    = "grey40"
    ),
    labels = c("Not significant", "Significant"),
    breaks = c(
      "Not significant",
      "Significant"
    )
  ) +

  guides(color = guide_legend(nrow = 1)) +

  labs(
    x = "Odds ratio (log scale)",
```

```r
    y = NULL
  ) +

  theme_light() +
  theme(
    panel.background  = element_rect(fill = "white", color = NA),
    panel.border      = element_blank(),
    panel.grid.major.x = element_blank(),
    panel.grid.minor.x = element_blank(),
    panel.grid.major.y = element_line(color = "grey85", linewidth = 0.4),
    panel.grid.minor.y = element_blank(),

    axis.line   = element_blank(),
    axis.ticks  = element_blank(),
    # x title no longer bold
    axis.title.x = element_text(size = 12, face = "plain"),
    axis.text   = element_text(size = 10, color = "black"),
    axis.text.y = element_text(size = 10, hjust = 0),

    legend.position = "top",
    legend.title    = element_blank(),
    legend.text     = element_text(size = 9),
    legend.key      = element_blank(),

    plot.background = element_rect(fill = "white", color = NA),
    plot.margin     = margin(2, 2, 2, 2)
  )
```

```r
# Get predictions for unadjusted and adjusted models
p1 <- as.data.frame(Predict(fit_unadjusted, hour_cat, fun = plogis)) %>%
  mutate(model = "Unadjusted")

p2 <- as.data.frame(Predict(fit_final_basic, hour_cat, fun = plogis)) %>%
  mutate(model = "Adjusted")

combined_plot <- bind_rows(p1, p2)
combined_plot$hour_cat_plot <- factor(
  combined_plot$hour_cat,
  levels = c("6-8am", "8-11am", "11-1pm", "1-3pm", "3-5pm", "5-7pm")
)
library(scales)  # for label_percent()

# Ensure hour_cat is a factor in the combined data too

ggplot(
  combined_plot,
  aes(x = hour_cat_plot, y = yhat, color = model, group = model)
) +
  # 95% CI ribbons
  geom_ribbon(
    aes(ymin = lower, ymax = upper, fill = model),
    alpha = 0.20,
    color = NA
```

```r
) +
# Lines for predicted probabilities
geom_line(size = 1.3) +

labs(
  x = "Surgery hour category",
  y = "Predicted 30-day mortality (%)"
) +

# Colors for unadjusted vs adjusted
scale_color_manual(
  name = NULL,
  values = c(
    "Unadjusted" = "#0BA6DF",
    "Adjusted"   = "#CC3311"
  )
) +
scale_fill_manual(
  name = NULL,
  values = c(
    "Unadjusted" = "#0BA6DF",
    "Adjusted"   = "#CC3311"
  )
) +

# Show y-axis as percentages (but keep underlying data as probabilities)
scale_y_continuous(
  labels = label_percent(accuracy = 0.01),  # e.g. 0.45%
  expand = expansion(mult = c(0, 0.05))
) +

# JAMA-ish theme
theme_bw() +
theme(
  panel.background   = element_rect(fill = "white", color = NA),
  panel.border       = element_blank(),
  panel.grid.major.x = element_blank(),
  panel.grid.minor.x = element_blank(),
  panel.grid.minor.y = element_blank(),
  panel.grid.major.y = element_line(color = "grey85", linewidth = 0.4),

  axis.line    = element_blank(),
  axis.ticks   = element_blank(),
  axis.title.x = element_text(size = 12),
  axis.title.y = element_text(size = 12),
  axis.text.x  = element_text(size = 10),
  axis.text.y  = element_text(size = 10),

  legend.position = "top",
  legend.title    = element_blank(),
  legend.key      = element_blank(),
  legend.text     = element_text(size = 10),
```

```r
    plot.background = element_rect(fill = "white", color = NA),
    plot.margin     = margin(5, 5, 5, 5)
  )
```

```r
#effect plot mortality_rsi
df_rsi <- as.data.frame(
  Predict(fit_final_basic, mortality_rsi, fun = plogis)
) %>%
  mutate(
    mort_pct = yhat * 100,
    lower_pct = lower * 100,
    upper_pct = upper * 100
  )

ggplot(df_rsi, aes(x = mortality_rsi, y = mort_pct)) +
  geom_line(size = 1.2, color = "#0BA6DF") +
  geom_ribbon(aes(ymin = lower_pct, ymax = upper_pct),
              alpha = 0.20, fill = "#0BA6DF") +
  labs(
    x = "Risk Stratification Index (30-day Mortality)",
    y = "Predicted 30-day Mortality (%)"
  ) +
  theme_bw() +
  theme(
    panel.grid.minor = element_blank(),
    axis.title = element_text(size = 12),
    axis.text = element_text(size = 10)
  )
```